

Unmasking Criminal Networks: AI-Based Graph Models Reveal Key Roles Criminals

Supplementary Material

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TABLE S1
DATA DICTIONARY FOR THE *INDIVIDUALS* DATASET.

Variable	Type	
nickname	categorical	0 = no criminal nickname; 1 = has criminal nickname
male	categorical	1 = male; 0 = not male
female	categorical	1 = female; 0 = not female
skin	categorical	0 = yellow; 1 = white; 2 = black; 3 = brown; 4 = other/unknown
hair	categorical	0 = curly; 1 = frizzy; 2 = straight; 3 = unknown
height	categorical	0 $\leq$ 1.50m; 1 = 1.50 - 1.60m; 2 = 1.61 - 1.70m; 3 = 1.71 - 1.80; 4 $\geq$ 1.80m; 5 = unknown
tattoo	categorical	0 = no prison tattoo; 1 = has prison tattoo
age	numeric	individual's age
weapon	categorical	0 = no weapon use; 1 = weapon use
arrested	categorical	0 = not arrested; 1 = arrested
convicted	categorical	0 = not convicted; 1 = convicted
rape	categorical	0 = did not commit; 1 = committed
extortion	categorical	0 = did not commit; 1 = committed
kidnapping	categorical	0 = did not commit; 1 = committed
theft	categorical	0 = did not commit; 1 = committed
homicide	categorical	0 = did not commit; 1 = committed
weapon trafficking	categorical	0 = did not commit; 1 = committed
drug trafficking	categorical	0 = did not commit; 1 = committed
faction	categorical	0 = does not belong; 1 = belongs
Dc	numeric	degree centrality
Bc	numeric	betweenness centrality
Cc	numeric	closeness centrality
CC	numeric	clustering coefficient

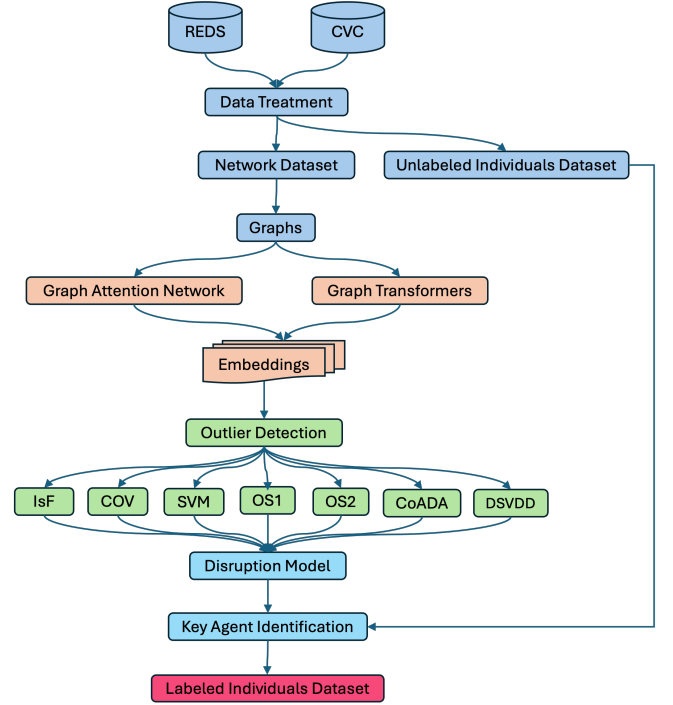


Fig. S1. Overview of the GRINDA framework. The pipeline integrates heterogeneous data, generates graph-based embeddings using Graph Attention Networks and Graph Transformers, applies multiple anomaly detection methods, and simulates network disruption to identify structurally critical individuals. The output is a labeled dataset for downstream analysis.

TABLE S2
EXECUTION TIME COMPARISON BETWEEN APPLE SILICON M4 AND GOOGLE COLAB G4

Nodes	Apple M4 (s)		Colab G4 (s)		Reduction (%)	
	GAT	GT	GAT	GT	GAT	GT
4053	94769.8	106801.9	53254.4	59275.1	43.8	44.5
1856	7706.4	8572.6	4415.6	4868.2	42.7	43.2
583	2084.3	2163.3	1227.6	1259.0	41.1	41.8
281	406.2	407.6	243.7	242.2	40.0	40.6
272	395.0	395.6	238.9	236.1	39.5	40.3
203	287.3	287.4	174.7	173.3	39.2	39.7
119	171.2	171.5	105.6	104.9	38.3	38.8
109	156.6	158.6	97.4	97.1	37.8	38.8
98	142.5	143.1	89.1	88.2	37.5	38.4
90	132.7	133.8	82.5	82.8	37.8	38.1
89	130.9	130.9	81.7	81.2	37.6	38.0
87	128.9	129.5	80.4	80.1	37.6	38.1
85	124.9	125.6	78.1	77.9	37.5	38.0
75	111.0	111.7	69.1	68.8	37.7	38.4
69	103.7	104.2	65.2	65.0	37.1	37.6
67	101.3	103.6	63.7	64.0	37.1	38.2
62	96.4	98.3	61.1	61.0	36.6	38.0
58	90.1	92.0	57.1	56.7	36.6	38.4
56	86.6	87.9	55.1	54.5	36.4	38.0
55	85.2	87.0	54.2	55.0	36.4	36.8
54	85.6	86.9	54.2	54.2	36.7	37.6
52	82.4	83.2	52.2	51.8	36.7	37.8
51	81.0	82.2	51.0	50.8	37.0	38.1
47	75.8	76.2	48.8	48.0	35.6	37.0
45	71.3	73.9	45.8	47.2	35.7	36.1
44	70.9	71.6	45.5	45.3	35.8	36.8
42	69.1	72.4	44.4	46.1	35.8	36.4
41	68.2	68.3	43.9	43.3	35.6	36.6
39	66.3	66.8	42.7	42.2	35.5	36.9
38	64.1	64.1	41.7	40.6	35.0	36.7
37	62.3	64.3	40.4	40.6	35.2	36.9
35	60.6	61.2	39.5	39.7	34.8	35.2
34	58.5	60.2	38.3	38.4	34.5	36.2
33	55.7	56.0	36.6	36.0	34.3	35.7
32	55.2	56.5	36.2	36.0	34.5	36.2
31	55.2	56.4	36.1	36.0	34.7	36.2
30	52.4	54.8	34.6	35.8	34.1	34.7

TABLE S3
BEST RESULTS BY NETWORK.

Network	Architecture	Nodes	Dimension	Model	Steps
3	GT	4,053	32	OS2	1,727
2	GT	1,856	64	OS2	608
40	GAT	583	128	DSVDD	251
177	GAT	281	128	DSVDD	115
88	GAT	272	64	DSVDD	115
22	GAT	203	128	DSVDD	96
229	GAT	119	32	DSVDD	44
159	GAT	109	128	DSVDD	52
611	GAT	98	64	DSVDD	46
128	GAT	90	128	DSVDD	12
1298	GT	89	128	DSVDD	35
87	GAT	87	128	DSVDD	48
170	GAT	85	256	DSVDD	11
73	GT	75	64	COV	27
109	GAT	69	128	DSVDD	25
28	GAT	67	64	DSVDD	10
154	GT	67	64	COV	10
220	GAT	62	64	DSVDD	24
528	GT	58	32	CoADA	25
233	GAT	56	128	DSVDD	14
314	GAT	55	256	DSVDD	22
678	GAT	54	32	DSVDD	25
1272	GAT	52	32	DSVDD	16
345	GT	51	128	CoADA	20
139	GAT	51	32	OS2	23
454	GAT	47	64	DSVDD	12
53	GT	45	32	CoADA	18
9	GAT	45	128	DSVDD	8
118	GAT	44	64	DSVDD	13
429	GAT	42	256	DSVDD	18
1053	GAT	42	256	DSVDD	13
596	GT	41	256	CoADA	14
124	GT	39	128	COV	19
884	GAT	38	32	DSVDD	16
610	GAT	37	64	DSVDD	11
882	GAT	35	32	DSVDD	7
202	GAT	35	64	DSVDD	7
1165	GT	35	32	SVM	10
1558	GAT	35	32	DSVDD	14
180	GT	34	64	DSVDD	22
574	GT	34	32	COV	7
134	GAT	34	32	DSVDD	11
1663	GAT	34	256	COV	7
797	GAT	33	64	DSVDD	10
63	GAT	33	64	DSVDD	14
45	GAT	33	32	DSVDD	11
292	GT	32	128	COV	13
81	GAT	32	64	DSVDD	10
47	GAT	32	256	IsF	8
886	GT	31	64	DSVDD	16
15	GAT	30	256	OS2	12

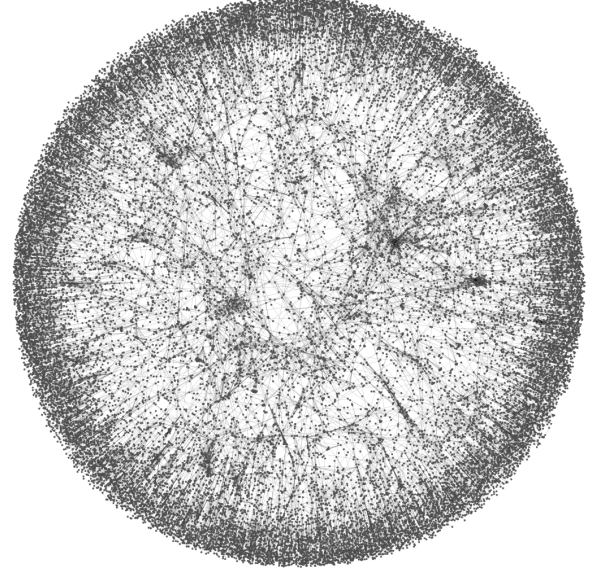


Fig. S2. Global criminal network from which the analyzed subnetworks are extracted. The graph contains 34,505 nodes (individuals) and 46,239 edges, representing diverse types of relationships recorded in the intelligence database. The data are continuously updated by intelligence agents based on field-collected information.

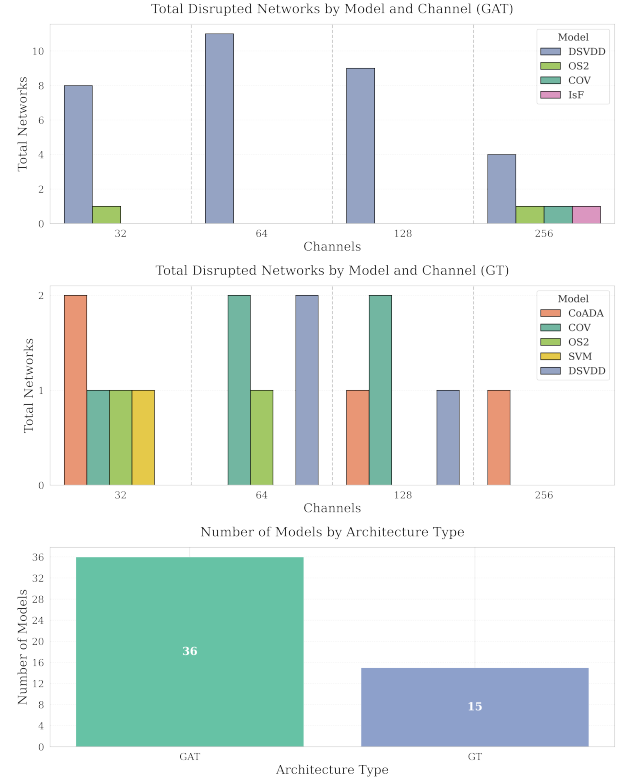


Fig. S3. Distribution of best-performing models across graph architectures (GAT and GT) and embedding dimensions. The figure reports the number of networks in which each model achieves the highest disruption, grouped by embedding dimension. The bottom panel summarizes the total number of selected models by architecture type, highlighting consistent selection patterns across network configurations.

TABLE S4

COMPARISON BETWEEN THE BEST GRINDA CONFIGURATION AND CLASSICAL CENTRALITY-BASED REMOVAL STRATEGIES. THE VALUES REPORTED FOR DEGREE CENTRALITY (Dc), BETWEENNESS CENTRALITY (Bc), AND CLOSENESS CENTRALITY (Cc) CORRESPOND TO THE NUMBER OF NODE REMOVALS (STEPS) REQUIRED TO ACHIEVE THE MAXIMUM DISRUPTION LEVEL OBTAINED BY THE BEST GRINDA CONFIGURATION IN EACH NETWORK. AVG. IMPROVEMENT REPRESENTS THE AVERAGE PERCENTAGE REDUCTION IN THE NUMBER OF REMOVALS REQUIRED BY GRINDA RELATIVE TO THE THREE CENTRALITY-BASED STRATEGIES.

Network	GRINDA	Dc	Bc	Cc	Avg. Improvement (%)
3	1727	1880	1845	1803	6.25
2	608	682	697	692	11.92
40	251	284	281	297	12.60
177	115	134	149	146	19.41
88	115	133	121	127	9.31
22	96	102	108	105	8.52
229	44	48	51	49	10.75
159	52	54	57	55	5.97
611	46	50	56	52	12.46
128	12	28	29	31	59.01
1298	35	43	45	44	20.42
87	48	61	64	62	22.96
170	11	17	18	17	36.49
73	27	37	39	38	28.92
109	25	36	38	41	34.60
28	10	16	17	19	42.02
154	10	26	25	26	61.03
220	24	35	37	36	33.30
528	25	36	38	37	32.40
233	14	29	30	29	52.26
314	22	34	36	37	38.24
678	25	36	38	37	32.40
1272	16	30	31	30	47.24
345	20	33	34	33	39.99
139	23	25	36	35	26.13
454	12	18	19	18	34.50
53	18	22	23	25	22.64
9	8	15	16	15	47.78
118	13	18	19	21	32.49
429	18	22	23	22	19.37
1053	13	18	19	21	32.49
596	14	19	20	25	33.44
124	19	22	24	23	17.29
884	16	20	21	20	21.27
610	11	17	18	17	36.49
882	7	14	15	13	49.83
202	7	13	15	14	49.83
1165	10	16	17	16	38.73
1558	14	19	20	21	29.88
180	22	24	26	25	11.90
574	7	14	15	13	49.83
134	11	17	18	17	36.49
1663	7	14	15	14	51.11
797	10	16	17	15	37.34
63	14	19	20	23	31.82
45	11	27	28	28	60.23
292	13	23	22	25	44.13
81	10	20	36	22	58.92
47	8	15	16	19	51.52
886	16	22	21	24	28.14
15	12	18	19	22	38.54
Mean Avg. Improvement					31.76